Siddharth Kumar

Aspiring GenAI Engineer — AI/ML — LLMs & RAG

Bengaluru, Karnataka, India | sidsiddhartha.2004@gmail.com | +91 63600 55019

linkedin.com/in/siddharth-kumar-01a1ba296 · github.com/siddharthskyhub

Professional Summary

Computer Science undergraduate and former Generative AI Intern at DRDO (CEMILAC) who builds practical, grounded LLM applications in Python. Hands-on with hybrid Retrieval-Augmented Generation (dense embeddings and BM25 fused with Reciprocal Rank Fusion), locally hosted LLMs through Ollama, and anti-hallucination guardrails that verify every generated claim against its source. Shipped two open-source projects, ResuMate AI and DocuMind, each with automated tests and a clean separation between deterministic logic and generative components. Prior experience as a Web Developer across three internships, delivering responsive front-end interfaces. Seeking a GenAI / AI-ML engineering role where reliable, explainable AI systems matter.

Technical Skills

Languages:	Python, JavaScript (ES6+), TypeScript, C/C++, SQL, Bash, HTML5/CSS3
AI, LLM & Agents:	Generative AI, LLMs (Llama 3.2, Mistral, Qwen, DeepSeek), Retrieval-Augmented Generation (RAG), Hybrid Search (BM25 + Dense), Reciprocal Rank Fusion, Semantic Search, Document Chunking, Embeddings (nomic-embed-text), Vector Databases (FAISS, ChromaDB), LangChain, LangGraph, AI Agents, Agentic Workflows, Prompt Engineering, Hallucination Guardrails, Citation Grounding, Ollama, Hugging Face
Machine Learning:	Supervised and Unsupervised Learning, Model Evaluation, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib
Backend & Systems:	FastAPI, RESTful APIs, Node.js, Express, Async I/O, Microservices Basics, Modular Architecture, File Parsing (PDF, DOCX)
Frontend & Full-Stack:	Streamlit, React, Next.js, Tailwind CSS, Responsive UI/UX, Session State Management
Databases:	PostgreSQL, MongoDB, SQLite, Data Modeling
Cloud, DevOps & Infra:	Docker, Git, GitHub, Linux/Unix, Streamlit Community Cloud, Vercel
Testing & Tools:	Pytest, Unit Testing, Test Fixtures, VS Code, Jupyter Notebook, Tableau (basics)
Core CS:	Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Professional Experience

DRDO, Ministry of Defence, Govt. of India – CEMILAC Jan 2026 – May 2026
Generative AI Intern *Bengaluru, India*

- Worked on Generative AI use cases at the Centre for Military Airworthiness & Certification, applying LLMs and Retrieval-Augmented Generation (RAG) to document-centric workflows.
- Built and tested Python-based prototypes demonstrating LLM question answering over technical documents, grounded in retrieved source passages.
- Experimented with prompt design, chunking, and retrieval strategies to improve the relevance and reliability of generated answers.
- Documented approaches and results, and demonstrated working prototypes to mentors and project stakeholders.

Technofly Solutions Jan 2025 – Feb 2025
Web Developer Intern *Bengaluru, India*

- Developed and maintained responsive front-end pages using HTML, CSS, and JavaScript.
- Collaborated with the team to implement features and fix UI issues across devices and browsers.
- Used Git for version control and followed team review practices before merging changes.

InternPe Aug 2024 – Sep 2024
Web Developer Intern *Remote*

- Built web application modules and delivered assigned projects within internship timelines.
- Practiced clean, reusable front-end code and version control with Git.

Teachnook (Teachscape Online Learning Services Pvt. Ltd.) Jul 2024 – Sep 2024
Web Developer Intern *Remote*

- Designed and developed web pages and small applications as part of a structured web development program.
- Applied responsive design principles and basic API integration in hands-on projects.

Key Projects

ResuMate AI – Privacy-First ATS Resume Optimizer [Live Demo] [GitHub]
Python, Streamlit, Ollama (Llama 3.2), nomic-embed-text, BM25, SQLite, Pytest

- Built a resume analysis platform that audits resumes against job descriptions and classifies each skill as Strong, Demonstrated, or Listed Only, based on evidence in context rather than keyword presence.
- Implemented hybrid RAG (dense embeddings + BM25) with Reciprocal Rank Fusion and a local Llama 3.2 model, using a strict grounding prompt with chunk IDs and a refusal policy for missing evidence.
- Engineered an anti-hallucination guardrail that verifies metrics, checks generated skills against source facts, and scores sentence faithfulness, rejecting and regenerating any output that invents numbers or tools; covered by Pytest unit tests.
- Added an ATS compliance checker (format, action verbs, contact details, parse rate), skill-gap analysis across up to 10 job descriptions, a SQLite Kanban application tracker, and a prerequisite-based learning roadmap.
- Designed a modular multi-page Streamlit app with separate deterministic engines and an optional AI layer; resumes are parsed fully in memory and the model runs locally, so no data leaves the device.

DocuMind – Hybrid Search RAG Engine for Technical Docs [GitHub]
FastAPI, Python, Ollama (Llama 3.2), nomic-embed-text, BM25, Pytest

- Built a RAG engine combining dense vector retrieval and BM25 through Reciprocal Rank Fusion, so exact queries such as error codes and function names are retrieved reliably where vector search alone fails.
- Implemented sliding-window paragraph chunking with metadata retention and a strict grounding prompt that attaches chunk references to every answer.
- Implemented a sentence-level citation verification guardrail that flags and rejects answer claims not supported by the retrieved document chunks.
- Developed FastAPI REST endpoints for uploading and indexing PDF, DOCX, TXT, and MD files, hybrid querying with tunable top-k, alpha, and temperature, document deletion, and health checks, with a web dashboard and Pytest suites covering chunking, BM25, RRF, and citation checks.

Education

Dayananda Sagar University, Bengaluru 2022 – 2026
Bachelor of Technology in Computer Science and Technology

Achievements & Certifications

- **DSU Code Red (24-hour hackathon)** – Qualified for the final round; organised by the DSU ACM Student Chapter, May 2024.
- **Hackfest 2024 (National-level hackathon, SAP-sponsored)** – Presented a solution in the regional round at New Horizon College of Engineering, May 2024.
- **Student Coordinator, Department Newsletter** – Contributed to the Department of Computer Science and Technology newsletter, Dayananda Sagar University (Dec 2023 – Dec 2025).
- **Technova Project Expo 2026** – Participated in the university project expo, Department of Computer Science and Technology, Apr 2026.
- Mastering Machine Learning: A Comprehensive Hands-On Bootcamp – CubeAISolutions Tech Pvt Ltd with Dayananda Sagar University, Dec 2024.
- Tata – GenAI Powered Data Analytics Job Simulation (Forage).
- Deloitte Australia – Technology Job Simulation (Forage).
- Introduction to Tableau – Simplilearn SkillUP, Oct 2025.